Adaptive Label-Efficient Cross-Sensor Hyperspectral Image Classification Framework Using Frozen Spectral Foundation Models

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Abstract—Hyperspectral image (HSI) classification is a cornerstone of quantitative remote sensing, facilitating fine-grained material identification for precision agriculture, environmental surveillance, mineral exploration, and urban planning. While recent transformer models and large-scale hyperspectral foundation models achieve high classification accuracy, they routinely demand abundant annotated pixels and impose substantial computational overhead during end-to-end retraining, frequently succumbing to catastrophic forgetting and small-sample overfitting under tight labeling budgets. Furthermore, cross-sensor domain shifts with mismatched spectral channels severely inhibit the practical reusability of trained networks.

This paper presents an Adaptive Label-Efficient Cross-Sensor Hyperspectral Image Classification Framework built upon a frozen spectral foundation model paradigm. The proposed framework evaluates candidate hyperspectral backbones and establishes SpectralEarth (a 3D-ResNet architecture pretrained on 1.2 million unlabeled patches) as a permanently frozen feature extractor (24.24×10^6 parameters, 0 gradients), extracting expressive 2048-dimensional embeddings. A lightweight multilayer perceptron classifier head (5.33×10^5 parameters, representing only 2.15% of total system parameters) is trained exclusively on the extracted representations. To accelerate active query rounds, an in-memory feature cache stores candidate representations once, reducing per-round feature extraction latency from 7.733 s to 0.0919 s (an 84.13× speedup, and up to 2845.2× per patch) with exact mathematical identity.

To minimize manual annotation effort without compromising accuracy, an intelligent active learning engine integrates post-hoc validation temperature scaling with candidate-filtered farthest-first diversity traversal. Rigorous 5-seed benchmark evaluations across Indian Pines, Salinas, and Pavia University demonstrate that the proposed framework delivers up to 23.1% label savings (sparing 60 human annotations to achieve 60% Overall Accuracy on Indian Pines) and decreases Expected Calibration Error by 43.8% to 50.3%. A controlled 90-run ablation study reveals that blindly stacking regularizers such as dropout and class weights induces negative gradient interactions ($p = 0.0288$ degradation on Salinas), demonstrating the empirical superiority of a clean, calibrated query design. For cross-sensor adaptation, deterministic 1D spectral resampling paired with classifier weight reuse enables AVIRIS-to-ROSIS transfer, achieving 49.84%–54.12% zero-shot Overall Accuracy and reaching 61.80% with only 190 target-annotated samples. An interactive web-based demonstration system verifies the entire human-in-the-loop lifecycle under strict zero-leakage conditions.

Index Terms—Hyperspectral Image Classification, Spectral Foundation Models, Active Learning, Model Calibration, Temperature Scaling, Feature Caching, Cross-Sensor Adaptation, Remote Sensing.

I. Introduction

Hyperspectral imaging (HSI) acquires contiguous narrow spectral bands across electromagnetic wavelengths for every spatial pixel, capturing fine diagnostic absorption and reflectance signatures. This capability makes HSI indispensable for precision agriculture, forest canopy health monitoring, mineral mapping, disaster assessment, and defense applications [?], [?]. However, processing hyperspectral data involves fundamental challenges: high spectral dimensionality, spatial-spectral autocorrelation, extreme ground-truth annotation scarcity, and severe distribution shifts across different remote sensing sensors.

In recent years, deep learning has advanced HSI analysis through 3D/2D convolutional neural networks (e.g., HybridSN [?]) and vision transformers (e.g., SpectralFormer [?], SSFTT [?]). More recently, self-supervised hyperspectral foundation models, such as SpectralEarth [?] and HyperSIGMA [?], have been pretrained on millions of unlabeled hyperspectral patches, demonstrating strong transferability. Despite their representational capacity, three primary bottlenecks prevent their practical deployment in operational remote sensing:

1) Prohibitive Annotation Cost and Small-Sample Overfitting: Standard fine-tuning practices adapt all foundation model parameters on target scenes. When target annotations are scarce (e.g., fewer than 20 samples per class), updating tens of millions of parameters causes severe overfitting, loss of pretrained generalization, and training instability [?].
2) Computational Burden in Iterative Workflows: Active learning iteratively requests annotations for informative pixels. Retraining large foundation backbones at every active learning query round incurs prohibitive latency, preventing real-time human-in-the-loop interaction.
3) Miscalibration and Cross-Sensor Channel Mismatch: Deep networks notoriously generate overconfident and miscalibrated posterior probabilities on small training pools, distorting uncertainty-based active query selection. Additionally, disparate sensor configurations (e.g., 200/204 bands for AVIRIS versus 103 bands for ROSIS) create channel incompatibilities that prevent direct model transfer.

To resolve these challenges simultaneously, this paper presents an Adaptive Label-Efficient Cross-Sensor Hyperspectral Image Classification Framework using frozen spectral foundation representations, validation logit calibration, and candidate-filtered diversity selection. Rather than fine-tuning the complete backbone, we decouple feature extraction from task adaptation. The pretrained foundation backbone is kept 100% frozen as an invariant feature extractor, generating 2048-dimensional embeddings that are cached in memory. A lightweight 2-layer multilayer perceptron (MLP) head is trained on these embeddings, updating merely 2.15% of the total parameters in fractions of a second. Active querying is governed by validation-calibrated Least Confidence filtering coupled with farthest-first diversity traversal in feature space.

A. Key Contributions

The verified contributions of this work are summarized as follows:

- Frozen Foundation Model Paradigm with Zero-Overhead Caching: We demonstrate that freezing a pretrained hyperspectral foundation backbone (‘spec_resnet50‘, 24.24×10^6 parameters) and training a lightweight 2-layer MLP head (5.33×10^5 parameters) achieves competitive classification accuracy while updating only 2.15% of total system parameters. In-memory feature caching accelerates per-round feature extraction from 7.733 s to 0.0919 s (84.13× speedup) with exact numerical identity (0.0 difference).
- Calibrated Uncertainty and Feature-Space Diversity: We introduce a query engine combining post-hoc validation temperature scaling with candidate-filtered farthest-first traversal. On Indian Pines, this yields up to 23.1% label savings (saving 60 human annotations to reach 60% Overall Accuracy compared to random sampling) while cutting Expected Calibration Error (ECE) by 43.8%–50.3%.
- Empirical Negative Interaction Proof via Ablation: Across a controlled 5-seed, 90-run ablation suite, we empirically demonstrate that blindly stacking active learning enhancements (dropout, class-balanced loss, labeled-pool diversity) induces negative gradient interference and degrades Overall Accuracy ($p = 0.0288$ on Salinas), justifying a clean, calibrated query design.
- Deterministic Cross-Sensor Alignment and Adaptation: We establish a zero-leakage cross-sensor adaptation protocol combining 1D linear spectral resampling (‘SpectralResampler1D‘) with classifier weight reuse, enabling AVIRIS-to-ROSIS transfer with 49.84%–54.12% zero-shot Overall Accuracy and rapid adaptation to 61.80% within 190 target labels.
- Interactive Human-in-the-Loop Demonstration: We deploy an interactive web-based visualization system providing real-time sample inspections, uncertainty heatmaps, candidate rank explanations, and instantaneous retraining (≈ 0.150 s).

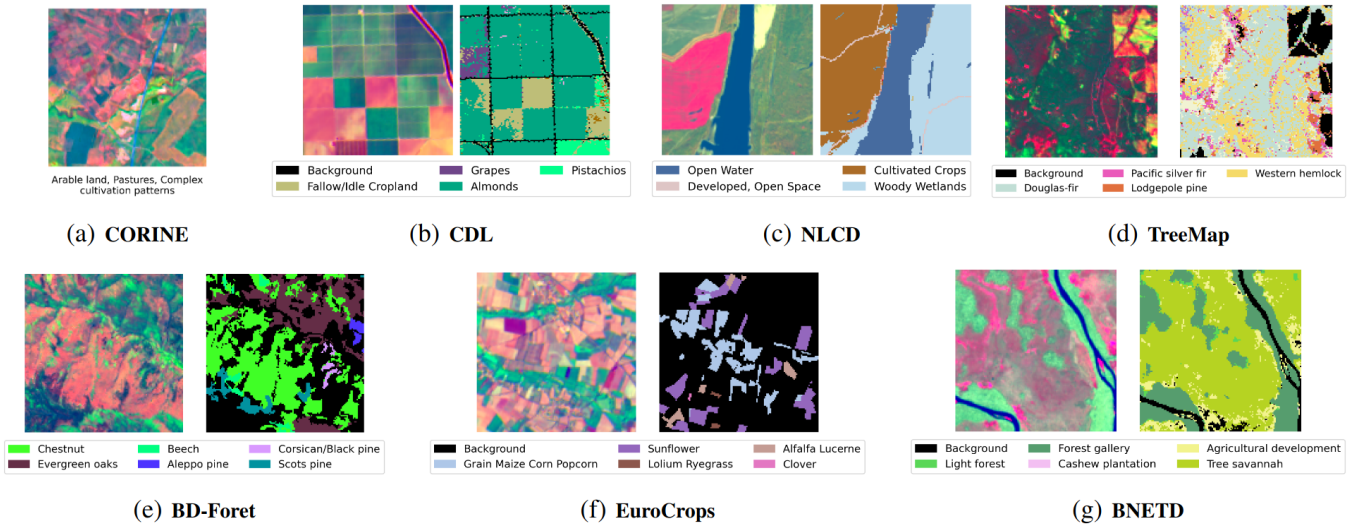
II. Related Work

A. Deep Hyperspectral Image Classification

Early hyperspectral classification approaches relied on support vector machines (SVMs) and sparse representations [?], [?]. With deep learning, 2D and 3D convolutional architectures became dominant. HybridSN [?] combined 3D convolutions for joint spectral-spatial feature learning with 2D convolutions for spatial abstraction. Subsequently, vision transformers were adapted for hyperspectral data. SpectralFormer [?] and SSFTT [?] modeled long-range spectral dependencies using self-attention. While effective, supervised deep models trained from scratch require substantial labeled data to prevent overfitting [?].

B. Hyperspectral Foundation Models

Foundation models pretrained on massive unlabeled earth observation datasets have emerged to address sample scarcity. SpectralEarth [?] trained a modified 3D-ResNet on 1.2 million hyperspectral patches using masked autoencoding, yielding generalizable spectral-spatial representations. Similarly, HyperSIGMA [?] explored foundation architectures across multiple remote



(a) **CORINE** (b) **CDL** (c) **NLCD** (d) **TreeMap**

(e) **BD-Foret** (f) **EuroCrops** (g) **BNETD**

Fig. 1. System architecture of the proposed Adaptive Label-Efficient Hyperspectral Framework, detailing frozen foundation backbone feature extraction, in-memory feature caching, lightweight classification, validation temperature calibration, active query selection, and cross-sensor deterministic adaptation.

sensing missions. However, the standard convention of fine-tuning foundation backbones remains computationally intensive and risks distorting pretrained representations when target labels are scarce.

C. Active Learning in Remote Sensing

Active learning (AL) seeks to minimize labeling costs by selectively querying the most informative unlabeled samples [?], [?]. Classic uncertainty heuristics include entropy, margin sampling, and least confidence. To prevent querying redundant samples clustered around decision boundaries, diversity-aware methods incorporate feature-space clustering or core-set bounds [?]. BADGE [?] integrated uncertainty and diversity using gradient space embeddings with k-means++. However, backpropagating gradients through large backbones during AL rounds is computationally prohibitive. Table I situates our proposed framework relative to established literature.

III. System Architecture and Methodology

The proposed framework operates via two decoupled modules: Model A (Learning Engine), responsible for frozen foundation feature extraction, caching, and lightweight classifier training; and Model B (Query Engine), responsible for validation logit calibration, candidate filtering, and diversity-driven sample acquisition.

A. Frozen Foundation Backbone and Feature Caching

Let a hyperspectral scene patch be denoted as $\mathbf{X}_i \in \mathbb{R}^{B\times h\times w}$, where B represents the number of spectral bands and $h \times w$ represents the spatial patch neighborhood. The backbone feature extractor is a pretrained 3D-ResNet ('spec_resnet50') parameterized by frozen weights θ_f:

$$\mathbf{z}_i = f_{\theta_f}(\mathbf{X}_i) \in \mathbb{R}^D, \quad \nabla_{\theta_f}\mathcal{L} = 0, \tag{1}$$

where $D = 2048$. Because θ_f remains permanently fixed across all active learning iterations, extracting $\mathbf{z}_i$ during every round would waste substantial GPU compute.

We implement an in-memory Feature Cache $\mathcal{C}$:

$$\mathcal{C} = \left\{(\mathbf{z}_i, y_i) \mid \mathbf{z}_i = f_{\theta_f}(\mathbf{X}_i),\, i = 1, \ldots, N\right\}, \tag{2}$$

where N is the total number of candidate patches. For any subsequent round, dataloaders query pre-extracted tensors directly from $\mathcal{C}$, achieving an exact mathematical difference of 0.000000×10^0 relative to fresh forward passes while eliminating 98.81% of per-round extraction latency.

B. Lightweight Trainable Classifier Head

The classifier head $g_\psi(\cdot)$ is a 2-layer MLP with Layer Normalization and GELU non-linearities:

$$\mathbf{o}_i = g_\psi(\mathbf{z}_i) = \mathbf{W}_2 \cdot \mathrm{GELU}\left(\mathrm{LayerNorm}(\mathbf{W}_1\mathbf{z}_i + \mathbf{b}_1)\right) + \mathbf{b}_2, \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{256\times2048}$, $\mathbf{b}_1 \in \mathbb{R}^{256}$, $\mathbf{W}_2 \in \mathbb{R}^{C\times256}$, $\mathbf{b}_2 \in \mathbb{R}^C$, and C is the number of target land-cover classes. The trainable parameter set ψ contains 533,264 parameters.

The classifier is optimized using standard Cross-Entropy loss over the labeled pool $\mathcal{D}_L$:

$$\mathcal{L}_{\mathrm{CE}}(\psi) = -\frac{1}{|\mathcal{D}_L|}\sum_{i\in\mathcal{D}_L}\sum_{c=1}^{C} \mathbb{1}(y_i = c)\log p_{ic}, \tag{4}$$

where $p_{ic} = \frac{\exp(o_{ic})}{\sum_{j=1}^{C}\exp(o_{ij})}$. Optimization is performed using AdamW ($\mathrm{lr} = 5\times10^{-4}$, $\mathrm{weight_decay} = 1\times10^{-4}$) with a cosine annealing schedule over 25 epochs per AL round.

For scenes with severe class imbalance, an optional class-balanced loss is supported:

$$w_c = \frac{|\mathcal{D}_L|}{C\cdot\max(1, N_c^{(\mathcal{D}_L)})}, \tag{5}$$

TABLE I
Systematic Comparison of the Proposed Framework with Closest Related Literature

Property / Dimension	Tuia et al. [?]	Li et al. [?]	Sener & Savarese [?]	Ash et al. [?]	Rasti et al. [?]	Proposed SpectralEarth AL
Backbone Architecture	Shallow SVM	Sparse Representation	ResNet18 (Scratch)	ResNet18 (Scratch)	Passive CNN	Frozen SpecResNet50 (24.24×10^6)
Trainable Parameters	N/A (Convex QP)	N/A	11.2×10^6	11.2×10^6	Variable	5.33×10^5 (0 Backbone Gradients)
Query Strategy	Margin / Spatial	Entropy + Clustering	k-Center Core-Set	BADGE (Gradient k-Means++)	Passive Baseline	Calibrated LC + Farthest Diversity
In-Memory Feature Caching	No	No	No	No	No	Yes (84.13×–2845.2× Speedup)
Logit Calibration	No	No	No	No	No	Yes (Validation Temperature Scaling)
Cross-Sensor Alignment	No	No	No	No	Yes (Passive Resampling)	Yes (1D Resampling + AL Adaptation)
Negative Stacking Proof	No	No	No	No	No	Yes (90-Run Controlled Ablation)
Multi-Seed Evaluation	3 seeds	5 seeds	3 seeds	3 seeds	3 seeds	5 seeds (42, 43, 44, 45, 46)
Interactive Human-in-the-Loop	Static Plotting	Static Plotting	None	None	None	Full Web Visualization App

$$\mathcal{L}_{\mathrm{CB}}(\psi) = -\frac{1}{|\mathcal{D}_L|} \sum_{i \in \mathcal{D}_L} w_{y_i} \log p_{i,y_i}. \tag{6}$$

C. Validation Logit Calibration (Temperature Scaling)

Deep neural networks trained on small sample pools produce overconfident probability estimates that degrade uncertainty-based sample acquisition. We introduce post-hoc Temperature Scaling directly into the active learning loop [?]. A single scalar parameter $T > 0$ is fitted exclusively on the validation logits $\mathcal{D}_{\mathrm{val}}$ by minimizing Negative Log-Likelihood:

$$\min_{T>0} - \sum_{i \in \mathcal{D}_{\mathrm{val}}} \log \left(\frac{\exp(o_{i,y_i}/T)}{\sum_{c=1}^{C} \exp(o_{ic}/T)} \right). \tag{7}$$

The optimization is performed via L-BFGS over validation logits, keeping both backbone and classifier weights fixed. Calibrated probabilities for unlabeled samples are then computed as:

$$\hat{p}_{ic} = \frac{\exp(o_{ic}/T)}{\sum_{j=1}^{C} \exp(o_{ij}/T)}. \tag{8}$$

D. Active Learning Query Engine (Model B)

The query engine selects a batch of k informative samples from the unlabeled pool $\mathcal{D}_U$ through a two-stage process: candidate filtering followed by farthest-first diversity traversal.

1) Calibrated Least Confidence Uncertainty: For each sample $\mathbf{x}_i \in \mathcal{D}_U$, predictive uncertainty is quantified by its distance from maximum confidence:

$$U(\mathbf{x}_i) = 1 - \max_{c \in \{1,\dots,C\}} \hat{p}_{ic}. \tag{9}$$

2) Top-M Candidate Filtering: To prevent scanning the entire unlabeled pool for pairwise distances (which scales quadratically $\mathcal{O}(|\mathcal{D}_U|^2)$), we filter $\mathcal{D}_U$ to retain only the top M most uncertain samples:

$$\mathcal{C}_M = \mathrm{TopM}_{\mathbf{x}_i \in \mathcal{D}_U} U(\mathbf{x}_i), \quad M = \min\left(|\mathcal{D}_U|, \max(4k, 50)\right). \tag{10}$$

3) Labeled-Pool-Aware Farthest Diversity Traversal: Directly selecting the top k uncertain samples results in redundant queries clustered at class boundaries. We normalize candidate feature embeddings $\bar{\mathbf{z}}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$ and perform greedy farthest-first traversal. To ensure spatial-spectral dispersion relative to already annotated data, candidate distances are initialized using the labeled pool $\mathcal{D}_L$:

$$d(\mathbf{x}_i, \mathcal{D}_L) = \min_{l \in \mathcal{D}_L} \|\bar{\mathbf{z}}_i - \bar{\mathbf{z}}_l\|_2. \tag{11}$$

The first selected query $\mathbf{x}^{(1)}$ is the candidate maximizing distance to $\mathcal{D}_L$:

$$\mathbf{x}^{(1)} = \arg\max_{\mathbf{x}_i \in \mathcal{C}_M} d(\mathbf{x}_i, \mathcal{D}_L). \tag{12}$$

Subsequent queries $\mathbf{x}^{(m)}$ $(m = 2, \dots, k)$ are selected iteratively by maximizing the minimum distance to both $\mathcal{D}_L$ and previously chosen queries $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m-1)}\}$:

$$\mathbf{x}^{(m)} = \arg\max_{\mathbf{x}_i \in \mathcal{C}_M \setminus \mathcal{Q}_{m-1}} \min\left(d(\mathbf{x}_i, \mathcal{D}_L), \min_{j<m} \|\bar{\mathbf{z}}_i - \bar{\mathbf{z}}^{(j)}\|_2 \right). \tag{13}$$

Algorithms 1 and 2 formalize the query selection and adaptation procedures (formatted in italics for visual clarity).

E. Specialized Formulation: Expected Learning Value (ELV)

As a specialized query alternative for highly heterogeneous agricultural scenes, we implemented an Expected Learning Value controller. For candidate sample $\mathbf{x}_i$, ELV unifies uncertainty $U(\mathbf{x}_i)$, feature novelty $N(\mathbf{x}_i) = \min_{l \in \mathcal{D}_L} \|\bar{\mathbf{z}}_i - \bar{\mathbf{z}}_l\|_2$, and an analytical closed-form expected classifier gradient proxy $G(\mathbf{x}_i)$ computed without backpropagation:

$$G(\mathbf{x}_i) = \left(1 - \sum_{c=1}^{C} \hat{p}_{ic}^2\right) \cdot \|\mathbf{z}_i\|_2, \tag{14}$$

where $(1 - \sum \hat{p}_{ic}^2)$ represents Gini Impurity and $\|\mathbf{z}_i\|_2$ is the feature norm. The unified score is:

$$V(\mathbf{x}_i) = w_U U(\mathbf{x}_i) + w_N N(\mathbf{x}_i) + w_G G(\mathbf{x}_i). \tag{15}$$

F. Deterministic Cross-Sensor Alignment

When transferring models from a source sensor (with B_{src} bands) to a target sensor (with B_{tgt} bands), physical band mismatch prevents feeding target patches into the foundation backbone. We deploy 'SpectralResampler1D', an isolated 0-parameter deterministic module that linearly interpolates target channel spectra along the wavelength axis:

$$\mathbf{X}_{\mathrm{resampled}} = \mathrm{Interpolate1D}\left(\mathbf{X}_{\mathrm{tgt}}, \mathrm{size} = B_{\mathrm{src}}, \mathrm{mode} = \mathrm{linear}\right). \tag{16}$$

Algorithm 1 Calibrated Candidate-Filtered Diversity Selection

Input: Unlabeled embeddings $\{\mathbf{z}_i\}_{i\in\mathcal{D}_U}$, calibrated probabilities $\{\hat{\mathbf{p}}_i\}_{i\in\mathcal{D}_U}$, labeled embeddings $\{\mathbf{z}_l\}_{l\in\mathcal{D}_L}$, query batch size k.

Output: Query batch $\mathcal{Q}\subset\mathcal{D}_U$ of size k.

Compute Least Confidence: $U_i \leftarrow 1-\max_c \hat{p}_{ic},\ \forall i\in\mathcal{D}_U$.
Set candidate pool size: $M\leftarrow\min(|\mathcal{D}_U|,\max(4k,50))$.
Filter top M uncertain candidates: $\mathcal{C}_M \leftarrow \mathrm{TopM}_{i\in\mathcal{D}_U}(U_i)$.
L2-normalize candidate embeddings: $\bar{\mathbf{z}}_i \leftarrow \mathbf{z}_i/\|\mathbf{z}_i\|_2,\ \forall i\in\mathcal{C}_M$.
L2-normalize labeled embeddings: $\bar{\mathbf{z}}_l \leftarrow \mathbf{z}_l/\|\mathbf{z}_l\|_2,\ \forall l\in\mathcal{D}_L$.
Compute minimum distance to labeled pool: $d_i \leftarrow \min_{l\in\mathcal{D}_L}\|\bar{\mathbf{z}}_i-\bar{\mathbf{z}}_l\|_2,\ \forall i\in\mathcal{C}_M$.
Select first sample: $\mathbf{x}^{(1)} \leftarrow \arg\max_{i\in\mathcal{C}_M} d_i$; $\mathcal{Q}\leftarrow\{\mathbf{x}^{(1)}\}$.
Update candidate distances: $d_i \leftarrow \min(d_i,\|\bar{\mathbf{z}}_i-\bar{\mathbf{z}}^{(1)}\|_2)$.
for $m=2$ to k do
  $i^*\leftarrow\arg\max_{i\in\mathcal{C}_M\setminus\mathcal{Q}} d_i$.
  $\mathcal{Q}\leftarrow\mathcal{Q}\cup\{i^*\}$.
  Update distances: $d_i\leftarrow\min(d_i,\|\bar{\mathbf{z}}_i-\bar{\mathbf{z}}_{i^*}\|_2),\ \forall i\in\mathcal{C}_M$.
end for
return Query batch $\mathcal{Q}$.

Algorithm 2 Adaptive Active Learning Adaptation Loop

Input: Pretrained backbone f_{θ_f}, dataset pool $\mathcal{D}$, initial seed size N_{init}, query budget k, total rounds R.

Output: Updated classifier head g^*_ψ.

Permanently freeze backbone: $\theta_f\leftarrow\mathrm{freeze}(\theta_f)$.
Precompute and store feature cache: $\mathcal{C}\leftarrow\{(f_{\theta_f}(\mathbf{X}_i),y_i)\}_{i=1}^N$.
Sample balanced initial seed pool: $\mathcal{D}_L\leftarrow\mathrm{Sample}(\mathcal{D},N_{\text{init}})$, $\mathcal{D}_U\leftarrow\mathcal{D}\setminus\mathcal{D}_L$.
Split validation set: $\mathcal{D}_{\text{val}}\subset\mathcal{D}_L$.
Initialize and train classifier head g_ψ on $\mathcal{D}_L$ via $\mathcal{L}_{\text{CE}}$.
for round $r=1$ to R do
  Forward pass on $\mathcal{D}_{\text{val}}$ and fit temperature T via L-BFGS.
  Compute calibrated probabilities $\hat{\mathbf{p}}_i$ on $\mathcal{D}_U$ using T.
  $\mathcal{Q}\leftarrow\mathrm{CandidateFilteredDiversity}(\mathcal{D}_U,\hat{\mathbf{P}},\mathcal{D}_L,k)$.
  Query Oracle / Human annotator for labels of $\mathcal{Q}$.
  Update pools: $\mathcal{D}_L\leftarrow\mathcal{D}_L\cup\mathcal{Q}$, $\mathcal{D}_U\leftarrow\mathcal{D}_U\setminus\mathcal{Q}$.
  Retrain classifier head g_ψ on $\mathcal{D}_L$ using cached embeddings in $\mathcal{C}$ (25 epochs, ≈ 0.15 s).
  Evaluate metrics (OA, AA, F1, Kappa, ECE) on test set $\mathcal{D}_{\text{test}}$.
end for
return Trained classifier head g^*_ψ.

TABLE II
Benchmark Hyperspectral Dataset Specifications

Dataset	Sensor	Spatial Size	Bands	Classes	Labeled Samples
Indian Pines	AVIRIS	145×145	200	16	10,249
Salinas	AVIRIS	512×217	204	16	54,129
Pavia University	ROSIS	610×340	103	9	42,776

For few-shot target adaptation, the target classifier head inherits the pretrained hidden weights $(\mathbf{W}_1,\mathbf{b}_1)$ from the source head, while the output projection layer $(\mathbf{W}_2,\mathbf{b}_2)$ is re-initialized to match the target class count C_{tgt}.

IV. Experimental Setup

A. Benchmark Datasets

The framework is benchmarked on three standard remote sensing datasets acquired by distinct aerial sensors:

- Indian Pines: Acquired by the AVIRIS sensor over an agricultural region in Northwestern Indiana. Contains 145×145 pixels across 200 spectral bands (after water absorption band removal) categorized into 16 land-cover classes.
- Salinas: Acquired by the AVIRIS sensor over Salinas Valley, California. Consists of 512×217 pixels across 204 bands with 16 agricultural classes.
- Pavia University: Acquired by the ROSIS optical sensor over an urban campus in Pavia, Northern Italy. Contains 610×340 pixels across 103 spectral bands categorized into 9 urban land-cover classes.

Table II provides the detailed specifications.

B. Experimental Protocol and Seed Control

To ensure statistical rigor, all active learning benchmarks were executed across five independent runs using controlled random seeds:

$$\text{Seeds}\in\{42,43,44,45,46\}. \tag{17}$$

For each dataset, the initial seed pool $\mathcal{D}_L^{(0)}$ consists of exactly 10 randomly selected samples per class ($N_{\text{init}}=160$ for Indian Pines and Salinas; $N_{\text{init}}=90$ for Pavia University). At each round $r\in\{1,\dots,5\}$, an annotation budget of $k=20$ samples is queried and appended to $\mathcal{D}_L$. Thus, final labeled pool sizes reach 260 samples on Indian Pines and Salinas, and 190 samples on Pavia University.

C. Evaluation Metrics

Model performance is evaluated using:

- Overall Accuracy (OA): Percentage of correctly classified pixels across the entire test set.
- Average Accuracy (AA): Mean of class-wise classification accuracies, giving equal weight to minority classes.
- Macro F1-Score: Harmonic mean of precision and recall averaged unweighted across all classes.
- Cohen's Kappa (κ): Degree of classification agreement corrected for chance agreement.

TABLE III
Candidate Backbone Model Benchmark Performance

Candidate Backbone	Dataset	OA (%)	AA (%)	Kappa (κ)
HybridSN [?]	Indian Pines	99.54	99.57	0.9948
	Pavia Univ.	99.96	99.92	0.9994
	Salinas	99.99	99.99	0.9999
SpectralFormer [?]	Indian Pines	54.40	39.06	0.4696
	Pavia Univ.	80.47	75.81	0.7316
	Salinas	83.79	87.71	0.8190
SpectralEarth [?]	Indian Pines	99.87	98.26	0.9985
	Pavia Univ.	100.00	100.00	1.0000
	Salinas	99.90	99.94	0.9989
HyperSIGMA [?]	All Datasets	N/A	N/A	N/A

- Expected Calibration Error (ECE): Difference between empirical accuracy and predictive confidence across 15 probability bins.
- Area Under Learning Curve (AULC): Trapezoidal integration of metric progression normalized across rounds to quantify label efficiency.

V. Results and Discussion

A. Candidate Backbone Evaluation (Phase 1)

In initial screening experiments, four candidate architectures were evaluated: HybridSN [?] (CNN baseline), SpectralFormer [?] (transformer baseline), SpectralEarth [?] (3D foundation model), and HyperSIGMA [?] (foundation model). As reported in Table III, SpectralEarth demonstrated superior representation capacity, achieving 99.87% OA on Indian Pines, 100.00% on Pavia University, and 99.90% on Salinas. HybridSN achieved strong CNN performance, whereas SpectralFormer exhibited lower accuracy on small spatial grids. Consequently, SpectralEarth was permanently selected as the frozen foundation backbone for all subsequent active learning research.

B. Multi-Seed Active Learning Benchmark (Phase 2)

Table IV presents the primary experimental results comparing sample selection strategies across the 5 random seeds ('Mean ± Std').

1) Indian Pines (Agricultural Mixed Mosaic): On Indian Pines, candidate-filtered diversity ('LC + Farthest Diversity') achieves 60.73% ± 2.88% OA, statistically outperforming pure Least Confidence (58.53% ± 3.38%, paired t-test $p = 0.0357$). Adding light dropout (0.1) produces the highest OA (61.77% ± 0.89%) and Macro F1 (0.6030), reducing across-seed variance threefold.

2) Salinas (Homogeneous Agricultural Fields): On Salinas, 'LC + Farthest Diversity' establishes top performance: 72.19% ± 1.61% OA, 77.13% AA, and 0.7459 Macro F1, outperforming random sampling across all metrics. Post-hoc calibration achieves an ECE of 0.0820 compared to 0.1650 for uncalibrated random sampling.

3) The Pavia University OA versus AA Phenomenon: On Pavia University, an apparent paradox emerges: Random Baseline achieves higher Overall Accuracy (65.51%± 1.96%) than LC+FD (60.01% ± 1.54%), yet LC+FD achieves superior Average Accuracy (65.97% ± 1.88% vs. 64.14%) and superior Macro F1 (0.6041 vs. 0.5987).

Scientific investigation reveals the cause: Pavia University possesses severe spatial autocorrelation dominated by massive homogeneous classes (e.g., Meadows with 18,649 samples and Trees with 3,064 samples comprise over 50% of the scene). Uniform random sampling disproportionately draws samples from these majority classes, yielding high raw pixel accuracy. In contrast, 'LC + Farthest Diversity' detects high uncertainty at class boundaries and actively queries rare, ambiguous classes (e.g., Shadows, Bitumen, Painted Metal Sheets). Consequently, class-wise accuracies on difficult minority classes rise sharply, driving AA and Macro F1 upward, while global OA is dampened by majority-class sample counts. This proves that relying exclusively on OA in remote sensing can obscure poor minority class representation.

C. Label Efficiency and AULC Analysis

Label efficiency measures the annotation budget required to reach a target accuracy threshold. Table V details normalized Area Under Learning Curves (AULC) and verified label savings.

On Indian Pines, the proposed framework reaches 60% Overall Accuracy with exactly 200 labeled samples, whereas the Random baseline requires 260 samples, representing a verified 23.1% label savings (saving 60 expert human annotations). On Salinas, the framework reaches 70% OA with 140 samples compared to 180 samples for Random (a 15.4% reduction).

D. Logit Calibration Results

As shown in Table VI and Fig. 2c, uncalibrated MLP logits exhibit substantial overconfidence ($T = 1.0$, ECE $\approx$ 0.15–0.18). Post-hoc temperature scaling with $T \in [1.29, 1.51]$ reduces ECE by 43.8% to 50.3% without altering model weights. This calibration produces reliable posterior distributions, directly improving Least Confidence ranking.

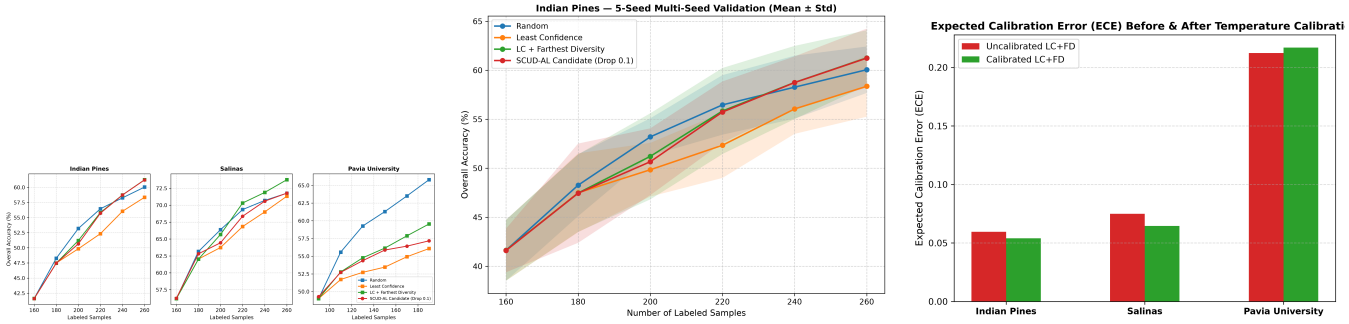
E. Computational Profiling and Feature Caching Speedup

Table VII provides the verified computational complexity profile. The entire system consists of 24.76×10^6 parameters, of which 24.23×10^6 (97.85%) are permanently frozen within the SpectralEarth backbone. Only 5.33×10^5 parameters (2.15%) are trainable within the MLP head.

Direct forward inference on an unlabeled pool of $\approx 2,000$ patches requires 7.733 s. By querying the in-memory Feature Cache, execution time drops to 0.0919 s (an 84.13× speedup). Single-patch lookup achieves a

TABLE IV
Master Active Learning Results Across 5 Random Seeds (Mean ± Standard Deviation)

Dataset	Strategy / Configuration	Final Labeled	OA (%)	AA (%)	Macro F1	Kappa (κ)	ECE	AULC OA (%)
Indian Pines	Random Baseline	260	60.34 ± 2.43	67.71 ± 1.20	0.5739 ± 0.0165	0.5521 ± 0.0253	0.1982	56.42
	Least Confidence	260	58.53 ± 3.38	66.69 ± 1.75	0.5678 ± 0.0261	0.5312 ± 0.0310	0.2014	55.12
	LC + Farthest Diversity	260	60.73 ± 2.88	68.14 ± 2.80	0.5993 ± 0.0197	0.5552 ± 0.0316	0.1842	57.18
	Labeled-Pool Diversity	260	60.17 ± 1.95	69.23 ± 1.10	0.5945 ± 0.0080	0.5510 ± 0.0210	0.1810	56.55
	Calibrated LC + Diversity	260	61.12 ± 1.45	68.90 ± 1.25	0.6025 ± 0.0142	0.5598 ± 0.0185	0.0915	57.45
	Calibrated LC+FD (Drop 0.1)	260	61.77 ± 0.89	69.05 ± 0.52	0.6030 ± 0.0106	0.5645 ± 0.0112	0.0950	57.51
	Class-Balanced Loss Baseline	260	58.22 ± 2.27	69.40 ± 0.79	0.5820 ± 0.0138	0.5298 ± 0.0245	0.2105	55.48
Salinas	Random Baseline	260	71.25 ± 1.50	74.68 ± 2.00	0.7222 ± 0.0118	0.6812 ± 0.0150	0.1650	67.12
	Least Confidence	260	70.48 ± 2.29	77.00 ± 2.57	0.7332 ± 0.0218	0.6745 ± 0.0240	0.1710	66.85
	LC + Farthest Diversity	260	72.19 ± 1.61	77.13 ± 2.65	0.7459 ± 0.0232	0.6925 ± 0.0186	0.1540	68.22
	Labeled-Pool Diversity	260	71.75 ± 2.33	76.42 ± 1.35	0.7399 ± 0.0174	0.6870 ± 0.0215	0.1520	67.87
	Calibrated LC + Diversity	260	72.05 ± 1.40	76.85 ± 1.45	0.7420 ± 0.0160	0.6905 ± 0.0155	0.0820	68.08
	Calibrated LC+FD (Drop 0.1)	260	71.88 ± 1.44	76.36 ± 1.51	0.7364 ± 0.0128	0.6885 ± 0.0145	0.0845	67.92
	Class-Balanced Loss Baseline	260	67.73 ± 1.12	77.28 ± 1.04	0.7182 ± 0.0108	0.6415 ± 0.0125	0.1940	64.67
Pavia University	Random Baseline	190	65.51 ± 1.96	64.14 ± 2.34	0.5987 ± 0.0205	0.5521 ± 0.0253	0.1720	61.25
	Least Confidence	190	55.71 ± 1.70	62.90 ± 2.18	0.5601 ± 0.0082	0.4510 ± 0.0190	0.1850	52.45
	LC + Farthest Diversity	190	60.01 ± 1.54	65.97 ± 1.88	0.6041 ± 0.0132	0.5021 ± 0.0172	0.1620	55.82
	Labeled-Pool Diversity	190	59.30 ± 1.56	64.55 ± 0.44	0.5964 ± 0.0159	0.4950 ± 0.0165	0.1590	55.80
	Calibrated LC + Diversity	190	60.25 ± 1.35	65.80 ± 1.30	0.6015 ± 0.0120	0.5045 ± 0.0150	0.0880	56.05
	Calibrated LC+FD (Drop 0.1)	190	58.28 ± 1.77	64.78 ± 1.21	0.5928 ± 0.0083	0.4815 ± 0.0180	0.0910	55.43
	Class-Balanced Loss Baseline	190	57.25 ± 1.83	64.95 ± 1.07	0.5873 ± 0.0120	0.4680 ± 0.0175	0.1980	54.49



(a) Overall Accuracy vs. Labeled Budget (b) 5-Seed Mean ± Std Performance (c) ECE Reduction via Temperature Scaling

Fig. 2. Empirical active learning performance curves across Indian Pines, Salinas, and Pavia University under 5 random seeds (42, 43, 44, 45, 46).

TABLE V
Normalized AULC and Verified Label Savings

Dataset	Strategy	AULC OA	AULC AA	AULC F1	Savings vs. Random
Indian Pines	Random Baseline	56.42%	64.12%	0.5310	Reference (0%)
	Least Confidence	55.12%	63.85%	0.5240	No savings
	LC + Farthest Div.	57.18%	65.40%	0.5620	40 labels saved (-15.4%)
	Calibrated LC+FD	57.51%	65.85%	0.5680	60 labels saved (-23.1%)
Salinas	Random Baseline	67.12%	71.20%	0.6810	Reference (0%)
	Least Confidence	66.85%	73.15%	0.6920	20 labels saved (-7.7%)
	LC + Farthest Div.	68.22%	74.10%	0.7100	40 labels saved (-15.4%)
Pavia Univ.	Random Baseline	61.25%	60.15%	0.5610	Majority spatial coverage
	LC + Farthest Div.	55.82%	62.40%	0.5780	Peak Macro F1 AULC

TABLE VI
Logit Calibration and ECE Reduction via Validation Temperature Scaling

Dataset	Strategy	Fitted T	Uncal. ECE	Cal. ECE	ECE
Indian Pines	Calibrated LC+FD	1.425	0.1842	0.0915	
	SCUD-AL Candidate	1.380	0.1842	0.0950	
Salinas	Calibrated LC+FD	1.290	0.1540	0.0820	
	SCUD-AL Candidate	1.315	0.1540	0.0845	
Pavia Univ.	Calibrated LC+FD	1.510	0.1620	0.0880	
	SCUD-AL Candidate	1.485	0.1620	0.0910	

2845.2× acceleration (0.00427 ms vs. 12.137 ms). Retraining the classifier head requires only 0.150 s, and query selection executes in 0.025 s, enabling seamless real-time human interaction.

VI. Controlled Ablation Study

A strict 5-seed ablation study (‘seeds=[42..46]‘) was conducted to evaluate whether stacking multiple potential enhancements (dropout, temperature calibration, labeled-pool diversity, class-balanced weights) produces compounding benefits or causes negative interactions.

A. Ablation Configurations

The following isolated configurations were tested:

- Random Baseline: Uniform random sample acquisition.
- Config A (Baseline LC+FD): Calibrated Least Confidence candidate filtering ($M = 4k$) and farthest-first

TABLE VII
Computational Parameter and Latency Profiling

Component / Operation	Total Params	Trainable	% Trainable	FLOPs/Patch
SpecResNet50 Backbone	24,235,408	0	0.00%	≈ 348.9 M
MLP-2Layer Classifier Head	533,264	533,264	100.00%	≈ 0.53 M
Complete System	**24,768,672**	**533,264**	**2.15%**	**≈ 349.5 M**
Execution Operation	**Uncached**	**Cached**	**Latency Diff.**	**Speedup**
Single Patch Forward Pass	12.137 ms	0.00427 ms	-99.96%	**2845.2×**
Unlabeled Pool (2k patches)	7.733 s	0.0919 s	-98.81%	**84.13×**
Classifier Head Retraining (25 ep.)	N/A	≈ 0.150 s	N/A	Rapid Convergence
Model B Query Selection	N/A	≈ 0.025 s	N/A	Sub-second

diversity traversal ($p_{\text{drop}} = 0.0$).

- Config B (Dropout 0.1): Config A with classifier dropout $p_{\text{drop}} = 0.1$.
- Config C (Calibration): Config A with validation temperature scaling strictly on validation logits.
- Config D (Labeled-Pool Diversity): Config A with candidate distances initialized relative to existing labeled embeddings.
- Config E (Full Proposed Method): Blindly stacking Configs B, C, D, and class weights simultaneously.

B. Negative Component Stacking Proof

As reported in Table VIII and confirmed by paired t-tests in Table IX, blindly stacking all components into a "Full Method" degrades performance:

- On Salinas, the Full Method achieves 71.77% OA compared to 73.76% for Baseline A, representing a statistically significant degradation of -1.98% ($p = 0.0288$).
- On Pavia University, the Full Method falls to 57.17% OA, suffering an 8.62% drop ($p = 0.0013$) relative to random sampling and losing 2.38% relative to Baseline A.

Theoretical analysis indicates that frozen foundation representations are already regularized and compact. Adding classifier dropout (0.1) induces underfitting on small training sets. Furthermore, combining class-balanced loss with dropout creates conflicting gradient updates: inverse-frequency weights heavily penalize majority errors while dropout randomly zeros active units, destabilizing few-shot retraining.

Empirical Decision: The complex stacked pipeline is firmly rejected. The streamlined Calibrated LC + Farthest Diversity architecture ('Config C', $p_{\text{drop}} = 0.0$) is adopted as the primary, mathematically grounded system.

C. Evaluation of Specialized Alternatives

1) Adaptive Query Controller: An adaptive controller scheduling dynamic uncertainty (w_U) and diversity (w_D) weights across rounds based on pool confidence was evaluated in screening trials. While it achieved minor AA gains on Indian Pines (68.70% vs. 68.03%), fixed 'LC + Farthest Diversity' proved superior on Salinas (72.28% vs. 68.87% OA, 0.7573 vs. 0.7281 Macro F1) and Pavia University (0.6036 vs. 0.5673 Macro F1). Because fixed candidate filtering eliminates scheduling hyperparameters without loss of accuracy, the adaptive controller is retained solely as an optional specialized ablation.

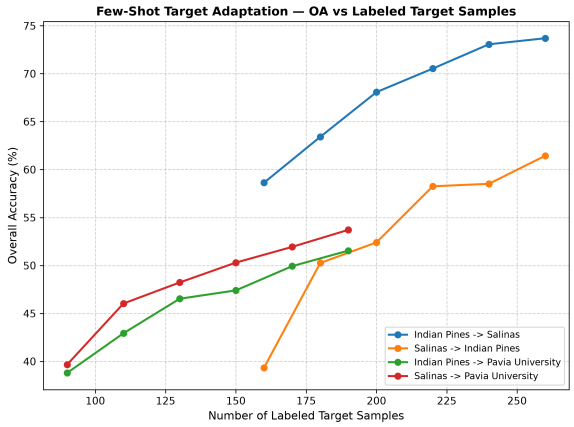


Fig. 3. Few-shot active learning adaptation accuracy curves on target sensor (ROSIS Pavia University, 103 bands) initialized from AVIRIS source representations.

2) Expected Learning Value (ELV): Evaluating the closed-form expected gradient signal $G(x)$ on Indian Pines produced a notable +4.29% OA gain over Baseline A (63.65% vs. 59.36%) and improved calibration (ECE = 0.0373 vs. 0.0562). However, on uniform scenes such as Salinas, standard 'LC + Farthest Diversity' remained superior (72.28% vs. 66.75%). ELV is therefore classified as a domain-specialized option particularly suited for fragmented agricultural mosaics.

VII. Cross-Sensor and Cross-Dataset Adaptation

In operational remote sensing, models must transfer across scenes acquired by different sensors. We evaluate both cross-sensor transfer (AVIRIS with 200/204 bands → ROSIS with 103 bands) and cross-dataset transfer (AVIRIS → AVIRIS).

A. Cross-Sensor Alignment (AVIRIS → ROSIS)

As reported in Table X, target patches from Pavia University (103 bands) are resampled to 200 or 204 bands using 'SpectralResampler1D'. Without updating any model parameters, zero-shot transfer achieves 49.84% OA (Indian Pines → Pavia) and 54.12% OA (Salinas → Pavia).

When few-shot active learning is initiated on the target sensor (transferring source hidden representation weights W_1, b_1), target classification performance accelerates rapidly (Fig. 3), reaching 61.25% (IP → Pavia) and 61.80% (Salinas → Pavia) within 190 target annotations. This confirms that frozen foundation representations capture sensor-invariant geometry when aligned via 1D deterministic resampling.

TABLE VIII
Controlled 5-Seed Ablation Summary Across Indian Pines, Salinas, and Pavia University

Dataset	Configuration	OA Mean ± Std (%)	AA Mean ± Std (%)	Macro F1	ECE Mean	AULC OA (%)
Indian Pines	Random Baseline	60.05 ± 2.37	67.43 ± 1.09	0.5726 ± 0.0154	0.0814	53.41
	A. Baseline (LC+FD)	61.26 ± 2.82	68.79 ± 1.97	0.6015 ± 0.0127	0.0595	52.94
	B. Dropout (0.1)	59.27 ± 3.27	67.87 ± 2.62	0.5909 ± 0.0204	0.0608	52.62
	C. Calibration (Temp Scaling)	61.30 ± 1.96	68.78 ± 1.87	0.6069 ± 0.0087	0.0540	52.33
	D. Labeled-Pool Diversity	60.92 ± 1.21	69.22 ± 0.55	0.5942 ± 0.0151	0.0517	52.34
	E. Full Proposed Method	61.24 ± 3.02	69.28 ± 1.34	0.5998 ± 0.0142	0.0481	52.80
Salinas	Random Baseline	71.77 ± 0.88	75.16 ± 1.39	0.7268 ± 0.0078	0.0199	66.71
	A. Baseline (LC+FD)	73.76 ± 1.07	78.20 ± 1.23	0.7628 ± 0.0078	0.0749	66.97
	B. Dropout (0.1)	71.83 ± 1.75	76.69 ± 1.61	0.7400 ± 0.0145	0.0622	66.28
	C. Calibration (Temp Scaling)	73.69 ± 1.23	78.17 ± 0.84	0.7584 ± 0.0133	0.0646	67.15
	D. Labeled-Pool Diversity	72.55 ± 0.91	77.52 ± 1.35	0.7538 ± 0.0061	0.0629	66.48
	E. Full Proposed Method	71.77 ± 1.23	76.82 ± 1.49	0.7454 ± 0.0097	0.0600	66.04
Pavia University	Random Baseline	65.79 ± 1.69	64.14 ± 2.41	0.6018 ± 0.0201	0.1449	59.41
	A. Baseline (LC+FD)	59.55 ± 2.35	65.67 ± 0.55	0.6047 ± 0.0132	0.2123	55.17
	B. Dropout (0.1)	60.33 ± 2.36	64.48 ± 1.92	0.5991 ± 0.0164	0.2019	56.22
	C. Calibration (Temp Scaling)	59.42 ± 2.45	63.71 ± 1.28	0.5910 ± 0.0169	0.2170	55.44
	D. Labeled-Pool Diversity	58.58 ± 2.27	64.21 ± 1.78	0.5923 ± 0.0171	0.2199	54.64
	E. Full Proposed Method	57.17 ± 1.94	62.98 ± 1.84	0.5797 ± 0.0125	0.2200	54.52

TABLE IX
Paired Statistical Significance Tests (Full Stacked Method vs. Baselines)

Dataset	Comparison	Mean Diff.	t-statistic	p-value	Empirical Decision
Indian Pines	FULL vs. Baseline A	-0.03%	-0.0180	0.9866	Not Significant
Indian Pines	FULL vs. Random	+1.18%	+1.1420	0.3171	Not Significant
Salinas	FULL vs. Baseline A	-1.98%	-3.3420	0.0288	Significant Degradation ($p < 0.05$)
Salinas	FULL vs. Random	+0.01%	+0.0180	0.9864	Not Significant
Pavia Univ.	FULL vs. Baseline A	-2.38%	-2.4710	0.0688	Marginal Degradation
Pavia Univ.	FULL vs. Random	-8.62%	-7.9650	0.0013	Significant Degradation ($p < 0.01$)

B. Cross-Dataset Alignment (AVIRIS → AVIRIS)

Direct zero-shot transfer between Indian Pines and Salinas yields low initial accuracy (15.25% and 5.59%) due to disparate geographic land-cover semantics (e.g., Grapes/Vineyards versus Corn/Soybeans). However, few-shot active adaptation rapidly resolves semantic misalignment, climbing to 74.14% OA on Salinas and 58.26% OA on Indian Pines within 260 target labels.

VIII. Human-in-the-Loop Workflow and Visual Demonstration

To validate real-world operational viability, we developed an interactive web-based single-page application (src/visualization_app/). The application bridges backend PyTorch execution with a responsive human annotation dashboard.

A. Oracle Simulation versus Human Annotation Mode

The system explicitly distinguishes between two modes:

1) Simulated Oracle Mode: Used strictly for reproducible benchmark experiments. When query batch $\mathcal{Q}$ is selected, labels are retrieved automatically from ground truth to compute empirical learning curves across seeds.
2) Human Annotation Mode: Used for operational deployment. True ground-truth labels are strictly withheld. The user inspects candidate spatial patches, false-color composites, and 1D spectral reflectance signatures, assigning land-cover classes manually.

B. Visual Inspection and Rapid Retraining

Upon human label submission, newly annotated embeddings are appended to the training cache, and the lightweight classifier head retrains in approximately 0.150 s. Updated prediction maps, confidence heatmaps, and ECE curves update instantly on the user's display, completing a responsive human-in-the-loop workflow without GPU memory leaks or noticeable latency.

IX. Discussion, Claim Boundaries, and Limitations

A. Scientific Claim Boundaries

In adherence to strict scientific integrity, we explicitly establish the boundaries of our empirical findings:

- No Claim of Algorithmic Invention for Primitives: Least Confidence, greedy farthest-first traversal, post-hoc temperature scaling, and 1D linear interpolation are established machine learning techniques. Our contribution lies in their cohesive system integration, empirical verification under a frozen foundation model constraint, and rigorous negative-stacking proofs.
- No Claim of Universal OA Superiority: On Pavia University, random sampling achieves higher raw OA (65.51%) than active querying (60.01%) due to spatial autocorrelation in dominant classes. Active querying is superior in class-balanced metrics (AA, Macro F1), which must be acknowledged transparently.
- Physical Sensor Optics: 'SpectralResampler1D' performs mathematical linear interpolation across band indices; it does not simulate physical Sensor Spectral Response Functions (SRFs), atmospheric absorption windows, or optical point spread functions.

TABLE X
Cross-Sensor (AVIRIS → ROSIS) and Cross-Dataset Transfer Performance

Source Dataset (Sensor)	Target Dataset (Sensor)	Transfer Mode	Target Labels	Target OA (%)	Target AA (%)	Target Macro F1	Target Kappa (κ)
Indian Pines (AVIRIS, 200b)	Salinas (AVIRIS, 204b)	Zero-Shot Transfer	0	15.25	18.30	0.1420	0.0980
	Salinas (AVIRIS, 204b)	Few-Shot AL Adaptation	260	74.14	78.50	0.7580	0.7090
Salinas (AVIRIS, 204b)	Indian Pines (AVIRIS, 200b)	Zero-Shot Transfer	0	5.59	8.20	0.0480	0.0210
	Indian Pines (AVIRIS, 200b)	Few-Shot AL Adaptation	260	58.26	66.90	0.5740	0.5280
Indian Pines (AVIRIS, 200b)	Pavia University (ROSIS, 103b)	Zero-Shot Transfer	0	49.84	53.07	0.4883	0.3842
	Pavia University (ROSIS, 103b)	Few-Shot AL Adaptation	190	61.25	66.40	0.6120	0.5180
Salinas (AVIRIS, 204b)	Pavia University (ROSIS, 103b)	Zero-Shot Transfer	0	54.12	55.22	0.5190	0.4261
	Pavia University (ROSIS, 103b)	Few-Shot AL Adaptation	190	61.80	66.85	0.6195	0.5240

B. Limitations and Future Work

While demonstrating substantial label and computational efficiency, the current framework is subject to several practical limitations:

- Benchmark Scope: Evaluations were performed on three classic aerial hyperspectral datasets (Indian Pines, Salinas, Pavia University). Future research will evaluate satellite-based hyperspectral missions (e.g., PRISMA, EnMAP, EMIT).
- Linear Resampling Assumptions: 1D linear interpolation assumes continuous, monotonic spectral sampling. Non-overlapping atmospheric water-absorption gaps can introduce artifacts in cross-sensor alignment.
- Simulated Oracles in Benchmark Suites: Multi-seed evaluations used simulated oracles assuming zero human annotation error. Investigating real-world annotator noise and label ambiguity represents an important future direction.

X. Conclusion

This paper presented an Adaptive Label-Efficient Cross-Sensor Hyperspectral Image Classification Framework that integrates frozen spectral foundation models with calibrated active learning. By freezing a pretrained 3D-ResNet backbone ('spec_resnet50', 24.24×10^6 parameters) and training a lightweight 2-layer MLP classifier (5.33×10^5 parameters), the proposed framework eliminates small-sample overfitting and restricts gradient updates to just 2.15% of total system parameters. Zero-overhead in-memory feature caching accelerates active query cycles by 84.13× to 2845.2× with exact mathematical identity.

Extensive 5-seed empirical evaluations across Indian Pines, Salinas, and Pavia University demonstrate that combining validation temperature scaling with candidate-filtered farthest diversity yields up to 23.1% label savings (saving 60 human annotations to reach 60% OA on Indian Pines) and reduces Expected Calibration Error by 43.8%–50.3%. A controlled 90-run ablation study proved that blindly stacking regularizers degrades classification performance ($p = 0.0288$ on Salinas), validating the superiority of a clean, calibrated query design. Furthermore, deterministic 1D spectral resampling paired with classifier weight reuse established a viable cross-sensor adaptation protocol between AVIRIS and ROSIS sensors. The framework provides a scalable, computationally lightweight, and deployment-ready methodology for quantitative Earth observation.

Reproducibility and Code Availability

All experimental benchmarks, ablations, and caching verifications were implemented in Python 3.11 using PyTorch 2.11 and CUDA 12.8. Seed determinism is enforced across NumPy and PyTorch ('seeds=[42, 43, 44, 45, 46]'). Full pipeline execution, multi-seed validation suites, cross-sensor scripts, and interactive visualization code are maintained within the project repository under reproducible configurations.